

## AQI Pulse: Air Quality / Wildfire Dashboard

### Datasets and Source:

For this project, we plan to utilize these three free, publicly available datasets:

1. **NASA FIRMS (Fire Information for Resource Management System)** — <https://firms.modaps.eosdis.nasa.gov/api/> — This gives satellite-based detections of active fires and hotspots, updated every few hours. Each detection includes the latitude/longitude, a confidence score, and a timestamp.
2. **Open-Meteo Air Quality API** — <https://open-meteo.com/en/docs/air-quality-api> — Provides real-time and historical air quality data (AQI, PM2.5, PM10, ozone) for any location, and it's completely free with no API key needed.
3. **Open-Meteo Weather API** — <https://open-meteo.com/en/docs> — From the same provider, this gives real-time wind speed and direction for the same coordinates.

### Purpose:

Wildfire smoke has become a growing public health issue, and air quality can go from fine to hazardous in just a few hours during fire season. Most tools I've come across just show a single AQI number with no real context — they don't tell you why it's changing or which direction the smoke is heading. I think that's a real gap: this kind of information matters to a lot of people, from parents and schools to outdoor workers and anyone with respiratory issues, who need more than just a number to make decisions about their day. On a personal level, I'm also excited about this project because it lets me get hands-on with real-time environmental data and figure out how to meaningfully combine multiple live sources into one useful signal.

### Questions I Intend to Investigate:

- Is there a meaningful relationship between nearby fire activity, wind direction, and short-term changes in ground-level air quality?
- Can that relationship be used to reliably classify a location's air quality trend — worsening, stable, or improving — instead of just reporting a static reading?
- How much does the delay between satellite fire updates and actual on-the-ground AQI changes affect how accurate a real-time trend estimate can be?